

18.600: Lecture 34

Entropy

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Outline

Entropy

Noiseless coding theory

Conditional entropy

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What is entropy?

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- ▶ Familiar on some level to everyone who has studied chemistry or statistical physics.
- ▶ Kind of means amount of randomness or disorder.
- ▶ But can we give a mathematical definition? In particular, how do we define the entropy of a random variable?

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- ▶ In information theory it's quite common to use $\log$ to mean $\log_2$ instead of $\log_e$. We follow that convention in this lecture. In particular, this means that

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- ▶ Since there are 2^k values in S , it takes k “bits” to describe an element $x \in S$.
- ▶ Intuitively, could say that when we learn that $X = x$, we have learned $k = -\log P\{X = x\}$ “bits of information”.

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- ▶ If a random variable X takes values $x_1, x_2, \dots, x_n$ with positive probabilities $p_1, p_2, \dots, p_n$ then we define the **entropy** of X by

$$H(X) = \sum_{i=1}^n p_i (-\log p_i) = - \sum_{i=1}^n p_i \log p_i.$$

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- ▶ This can be interpreted as the expectation of $(-\log p_i)$. The value $(-\log p_i)$ is the “amount of surprise” when we see x_i .

Twenty questions with Harry

- ▶ Harry always thinks of one of the following animals:

x	$P\{X = x\}$	$-\log P\{X = x\}$
Dog	1/4	2
Cat	1/4	2
Cow	1/8	3
Pig	1/16	4
Squirrel	1/16	4
Mouse	1/16	4
Owl	1/16	4
Sloth	1/32	5
Hippo	1/32	5
Yak	1/32	5
Zebra	1/64	6
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- ▶ Can learn animal with $H(X) = \frac{47}{16}$ questions on average.

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- ▶ If X takes one value with probability 1, what is $H(X)$?
- ▶ If X takes k values with equal probability, what is $H(X)$?
- ▶ What is $H(X)$ if X is a geometric random variable with parameter $p = 1/2$?

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Coding values by bit sequences

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- ▶ If X takes four values A, B, C, D we can code them by:

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- ▶ No sequence in code is an extension of another.
- ▶ What does 100111110010 spell?
- ▶ A coding scheme is equivalent to a twenty questions strategy.

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- ▶ **Note:** The expected number of questions *is* the entropy if each question divides the space of possibilities exactly in half (measured by probability).
- ▶ In this case, let X take values $x_1, \dots, x_N$ with probabilities $p(x_1), \dots, p(x_N)$. Then if a valid coding of X assigns n_i bits to x_i , we have

$$\sum_{i=1}^N n_i p(x_i) \geq H(X) = - \sum_{i=1}^N p(x_i) \log p(x_i).$$

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- ▶ **Data compression:** $X_1, X_2, \dots, X_n$ be i.i.d. instances of X . Do there exist encoding schemes such that the expected number of bits required to encode the entire sequence is about $H(X)n$ (assuming n is sufficiently large)?

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- ▶ Yes. We can cut space of N^n possibilities *close* to exactly in half at each stage (up till near end maybe).

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- ▶ Claim: if X and Y are independent, then

$$H(X, Y) = H(X) + H(Y).$$

Why is that?

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- ▶ We can define a **conditional entropy** of X *given* $Y = y_j$ by

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- ▶ We similarly define $H_Y(X) = \sum_j H_{Y=y_j}(X)p_Y(y_j)$. This is the *expected* amount of conditional entropy that there will be in Y after we have observed X .

Properties of conditional entropy

- Definitions: $H_{Y=y_j}(X) = -\sum_i p(x_i|y_j) \log p(x_i|y_j)$ and $H_Y(X) = \sum_j H_{Y=y_j}(X)p_Y(y_j)$.

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- ▶ To prove this property, recall that $p(x_i, y_j) = p_Y(y_j)p(x_i|y_j)$.
- ▶ Thus,
$$\begin{aligned} H(X, Y) &= -\sum_i \sum_j p(x_i, y_j) \log p(x_i, y_j) = \\ &= -\sum_i \sum_j p_Y(y_j)p(x_i|y_j) [\log p_Y(y_j) + \log p(x_i|y_j)] = \\ &= -\sum_j p_Y(y_j) \log p_Y(y_j) \sum_i p(x_i|y_j) - \\ &\quad \sum_j p_Y(y_j) \sum_i p(x_i|y_j) \log p(x_i|y_j) = H(Y) + H_Y(X). \end{aligned}$$

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- ▶ In words, the expected amount of information we learn when discovering X *after* having discovered Y can't be more than the expected amount of information we would learn when discovering X *before* knowing anything about Y .
- ▶ Proof: note that $\mathcal{E}(p_1, p_2, \dots, p_n) := -\sum p_i \log p_i$ is concave.
- ▶ The vector $v = \{p_X(x_1), p_X(x_2), \dots, p_X(x_n)\}$ is a weighted average of vectors $v_j := \{p_X(x_1|y_j), p_X(x_2|y_j), \dots, p_X(x_n|y_j)\}$ as j ranges over possible values. By (vector version of) Jensen's inequality,
$$H(X) = \mathcal{E}(v) = \mathcal{E}(\sum p_Y(y_j) v_j) \geq \sum p_Y(y_j) \mathcal{E}(v_j) = H_Y(X).$$